

- a. *Vowpal Wabbit*: Library services for machine learning to enable text analytics like sentiment analysis in tweets.
- b. *Cognitive services on Spark*: These combine the features of Microsoft cognitive services in SparkML pipelines in order to get a solution design for cognitive data modelling services like anomaly detection.
- c. *LightBGM*: Machine learning model to enable training for predictive analysis like face ID detection.
- d. *Spark Serving*: Provides a REST API facility to use Spark streaming resources at ease.
- e. *HTTP on Spark*: Enables distributed microservices orchestration in integrating Spark and HTTP protocol based accessibility.
- f. *Cognitive Toolkit (CNTK)*: Deep learning framework from Microsoft to enable a Spark solution for data validation in streaming data.
- g. *Lime*: Classifier and modelling solution for distributed models in Spark streaming data.
- h. *Spark Binding Auto-generation*: Enables PySpark (Python) and SparklyR (R programming) coding facility for modelling Spark streaming data models.

MMLSpark enables data processing with Spark for modelling solutions using machine learning libraries for image processing, video analytics, text analytics, etc.

Latest version: 4.5.5

Official website: <https://opencv.org/>

PyTorch

PyTorch is an open source machine learning framework based on the Torch library and is primarily used for computer vision and natural language processing (NLP). Developed by the Facebook AI Research Lab, it facilitates building deep learning models and allows them to be expressed in idiomatic Python. It is an optimised tensor library for deep learning using GPUs and CPUs. Functionality

can be extended with common Python libraries such as NumPy and SciPy. The easiest way to get started with PyTorch is to use NGC containers. The PyTorch NGC container comes with all the dependencies that are useful to design applications for conversational AI, natural language processing, recommendation systems and computer vision.

It has two good features:

- Tensor computations with strong GPU acceleration.
 - Deep neural networks are built on a tape-based autograd system.
- The PyTorch library comprises the following components:
- *Torch*: This is considered a tensor library like NumPy.
 - *Torch.autograd*: This is a tape based automatic differentiation library.
 - *Torch.jit*: Torch script to create serialisable and optimisable models from code.
 - *Torch.nn*: This is a neural network library.
 - *Torch.multiprocessing*: Python multiprocessing and useful for data loading and Hogwild training.
 - *Torch.utils*: Miscellaneous utilities for developers for using code.

Features:

- a. *Production ready*: Easy to use and has flexibility to integrate code in runtime environments.

- b. *Torchserve*: Strong tool for deploying PyTorch in an easy manner.
- c. *Mobile support*: Supports end-to-end workflow — from Python to deployment on iOS and Android environments.
- d. Robustness in integration of tools and libraries.
- e. With a C++ frontend, design and architecture becomes easy to understand, and has low latency.
- f. *Cloud computing ready*: PyTorch is well supported for cloud API-based seamless integration.

Latest version: 1.11.0

Website: <https://pytorch.org/>

OpenNN

OpenNN is a C++ based software library designed for advanced analytics and implements neural networks. The library is stable and very fast in memory allocation. It is constantly optimised and parallelised to attain efficiency in code. It implements data mining methods.

Features:

- a. High processing speed
- b. Easy design and implementation
- c. Proof-of-concept
- d. C++ oriented

Website: <https://www.opennn.net/> 

References

- [1] <https://azure.microsoft.com/en-in/services/cognitive-services>
- [2] <https://research.facebook.com/ai/>
- [3] <https://medium.com/senseai/deep-learning-101-role-of-deep-learning-in-artificial-intelligence-d949d0ffc4f6>
- [4] <https://ai.google/tools/>
- [5] <https://opencv.org/>
- [6] <https://pytorch.org>
- [7] <https://www.opennn.net/>

 **By: Dr Anand Nayyar and Dr Magesh Kasthuri**

Dr Anand Nayyar is a PhD in wireless sensor networks and swarms intelligence. He works at Duy Tan University, Vietnam, and loves to explore open source technologies, IoT, cloud computing, deep learning and cyber security.

Dr Magesh Kasthuri is a senior distinguished member of the technical staff and principal consultant at Wipro Ltd. This article expresses his views and not that of Wipro.